

Decision-Level Auditing of Latent Robot World Models

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Abstract—Latent world models are often judged by prediction error, but robot planning depends on whether predicted futures rank candidate actions correctly. We introduce a decision-level audit that separates these questions. On frozen 300-action Push-T menus from released JEPA-WM and DINO-WM planners, we compare the action chosen from predicted terminal latents (P) with the action chosen after scoring simulator-realized endpoints in the same latent space (O). We evaluate both choices using goal-distance utility G (negative final-state distance) and task reward J . Across 20 trajectory clusters and 562 model-matched rows, predicted scoring substantially outperformed uniform menu selection. Replacing predictions with realized endpoints, however, did not reliably improve physical outcomes: the raw O -versus- P effect was -5.30 in G (95% CI $[-9.64, -0.71]$, Holm-adjusted exact $p = .0755$) and $+0.042$ in J ($[-0.176, 0.240]$, $p = .701$). Across all 577 merge-valid diagnostic artifacts, P and O selected different actions in 337; only 48 and 38 of those switches reversed the strict G -best and J -best pairwise ordering, respectively. The central result is a separation: the latent scores were useful for choosing actions, but more faithful terminal latents did not reliably make those choices better. A post-hoc, prediction-only near-tie score nevertheless tracked these selector changes (exploratory AUROC .714, 95% trajectory-bootstrap CI $[.665, .762]$): the held-out top ambiguity quarter switched 82.4% of the time versus 51.0% otherwise. It did not identify higher planner regret, so we treat it only as a candidate trigger for prospective verification rather than call it physical risk or epistemic uncertainty.

I. INTRODUCTION

World models matter to robot planning through the decisions they induce. A planner can fail because its model predicts the wrong future, but it can also fail because its learned representation assigns a low cost to a physically poor outcome. Conversely, an imperfect predictor may still preserve the action ordering needed for control. Standard prediction losses do not separate these cases. Planning-oriented model learning and value equivalence formalize the distinction [1]–[4]; recent benchmarks measure world models through ranking and regret [5], [6].

We ask a direct counterfactual question: *if the same candidate actions are scored using what actually happened in the simulator, does the planner choose a better action?* For each candidate, we replace its predicted terminal latent with an encoding of its realized endpoint. The action menu, encoder, latent goal, cost function, and physical evaluation remain fixed. Any improvement is therefore decision quality recovered by removing terminal prediction error from the existing planning interface.

In plain terms, P asks what the planner chooses from its prediction, O asks what it chooses after seeing the realized

endpoint, and G/J judge both selected actions by their physical outcomes.

We make three contributions. First, we define a matched $P/O/G/J$ intervention that connects latent prediction errors to action choice and physical regret. Second, we derive exact candidate- and pair-level score accounting that explains which prediction errors change a finite-menu decision without requiring semantic coordinates or a global model of latent geometry. Third, we apply the audit to released JEPA-WM and DINO-WM Push-T planners under a preregistered analysis. The predicted scores contain useful action-selection signal, yet replacing predicted endpoints with realized ones does not recover decision quality on average. As a separate exploratory analysis, a selective diagnostic shows that the predicted top-two margin can flag selector instability before simulator replay, while failing to flag higher physical regret.

II. RELATION TO PRIOR EVALUATION

Decision-aware model learning distinguishes predictive accuracy from the quantities a downstream controller uses. Value-aware losses and value equivalence formalize models that preserve decisions without reconstructing every aspect of the environment [1], [2], [4], while goal-aware prediction emphasizes task-relevant features [3]. Recent benchmarks likewise evaluate world models through downstream ranking, regret, and control rather than reconstruction alone [5], [6].

Robot world-model studies have also isolated objective and interface failures. Latent objectives may discard information their planners need [7], and failures can remain hidden in aggregate offline metrics [8]. Our claim is narrower than proposing a new loss or benchmark suite. Unlike decision-fidelity benchmarks that score a model’s downstream utility, our audit intervenes within a fixed planner: it replaces only predicted terminal endpoints with their realized, re-encoded counterparts, estimating the decision regret recoverable specifically from terminal prediction correction. Unlike objective-swap studies, the representation, objective, candidate set, and planner interface remain fixed.

III. MATCHED DECISION INTERVENTION

A. Four views of the same action

For action sequence U_i , let $\hat{v}_i, \hat{r}_i$ denote its predicted visual and proprioceptive terminal latents, v_i, r_i the corresponding latents after simulator replay and frozen re-encoding, and q_v, q_r the encoded goal. The released lower-is-better costs are

$$\begin{aligned} P_i &= \text{MSE}(\hat{v}_i, q_v) + \alpha \text{MSE}(\hat{r}_i, q_r), \\ O_i &= \text{MSE}(v_i, q_v) + \alpha \text{MSE}(r_i, q_r), \end{aligned} \quad (1)$$

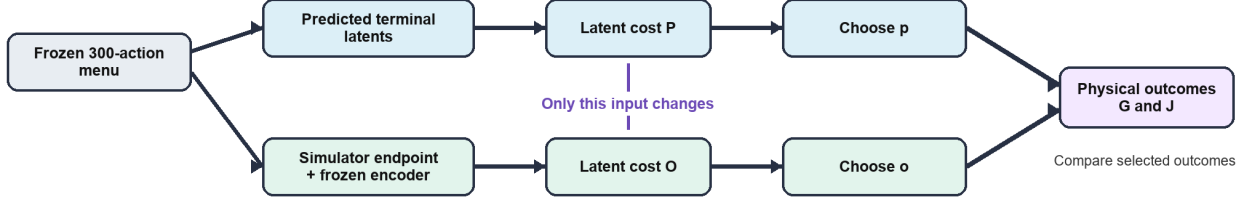


Fig. 1. Matched decision intervention. The candidate actions, encoder, goal latent, and cost are fixed. Only the terminal state supplied to the latent cost changes: model prediction for P , simulator-realized endpoint for O . Native physical outcomes G and J evaluate the two selected actions.

with the released proprioceptive weight $\alpha = 0.1$. G_i is negative official final simulator state distance to the dataset goal, and J_i is the float32 sum of the 30 official Push-T coverage rewards. Thus P/O are lower-is-better latent costs, whereas G/J are higher-is-better physical utilities. Crucially, O is not a physical oracle: it supplies the true endpoint to the same potentially misaligned learned encoder and latent objective. With $p = \arg \min_i P_i$, $o = \arg \min_i O_i$, and $Y \in \{G, J\}$, define

$$\Delta^Y = [\max_i Y_i - Y_p] - [\max_i Y_i - Y_o] = Y_o - Y_p. \quad (2)$$

Positive Δ^Y favors O over P . At the aggregate level, a non-positive estimate indicates no average recovery under the unchanged interface.

This estimand is finite-menu and action-matched. Let $R_s^Y = \max_i Y_i - Y_s$ be the regret of selector $s \in \{p, o\}$. Then $\Delta^Y = R_p^Y - R_o^Y$ measures regret removed by terminal prediction correction, whereas R_o^Y is regret that remains after correction. The latter can arise from the representation, latent goal cost, terminal-only scoring, or candidate support; the intervention does not identify those causes separately.

B. Exact latent-score accounting

For either latent block, let $q \in \mathbb{R}^d$ be the goal, z_i the encoded realized endpoint, and $\hat{z}_i = z_i + e_i$ its prediction. Expanding the square gives

$$\begin{aligned} P_i - O_i &= d^{-1} \|\hat{z}_i - q\|_2^2 - d^{-1} \|z_i - q\|_2^2 \\ &= \underbrace{2d^{-1} (z_i - q)^\top e_i}_{D_i} + \underbrace{d^{-1} \|e_i\|_2^2}_{M_i}. \end{aligned} \quad (3)$$

D_i is signed goal-directional error: it records whether error moves the prediction toward or away from the latent goal. $M_i \geq 0$ is error magnitude. We compute both terms separately for visual and proprioceptive blocks and combine them with the same $\alpha = 0.1$ as Eq. 1. For candidates a, b , the change in their score margin is exact:

$$(P_a - P_b) - (O_a - O_b) = (D_a - D_b) + (M_a - M_b). \quad (4)$$

Thus the audit can attribute a ranking reversal to directional or magnitude terms without interpreting individual coordinates. This is local score accounting, not a claim that the latent space is globally identifiable or physically disentangled.

C. What the intervention can establish

Three outcomes are possible. If P and O agree, prediction error did not change the selected action on that menu. If they disagree and $\Delta^Y > 0$, correction recovered some physical regret. If they disagree without positive aggregate recovery, prediction error changed the planner's ranking but did not provide evidence that it was a sufficient or dominant recoverable bottleneck under the frozen representation and interface. The last case motivates diagnosis of the remaining interface; it does not by itself prove which component is faulty.

IV. FROZEN PUSH-T STUDY

A. Models, task, and candidate support

We audit released JEPA-WM and DINO-WM Push-T checkpoints [9], [10]. Both use DINOv2 features, so this is not a comparison of independent representation families. Each issued row holds an outcome-blind common-initial menu of 300 six-step actions matching released CEM initialization. Every action is replayed from the same state. The audit covers iteration-0 menus, not later CEM populations or closed-loop control.

Push-T requires a planar agent to push a T-shaped block toward a target pose. It is useful here because different short action sequences can look similar in latent cost while producing different block displacement and task reward. For each of 20 validation trajectories, the study samples five outcome-blind offsets and three menu seeds for both predictors: $20 \times 5 \times 3 \times 2 = 600$ issued model rows. Candidate actions are identical across the P , O , G , and J views within a row, eliminating candidate-generation changes as an explanation for a within-row selector difference.

B. Frozen inference and validity rules

Trajectories are the inferential units. A cell is one trajectory–offset–seed combination containing both model rows. The frozen analysis uses 20 trajectory clusters, a model-matched mask (562 valid model rows; 281 matched cells), and a fixed 15-cell denominator per trajectory (5 offsets $\times$ 3 seeds). Within each cell, the two models receive equal weight; cell effects are averaged within trajectory before inference. The analysis uses deterministic 10,000-resample

percentile bootstrap intervals, and exact trajectory-level sign-flip tests (assuming sign symmetry). Holm correction is applied separately within each preregistered two-outcome G/J family for raw, bounded- S , and P -versus-uniform analyses. Invalid or unmatched cells contribute zero *evidence* for both models in the fixed denominator, not a physical outcome of zero. This fixed-denominator rule preserves equal scheduled exposure rather than allowing whichever model completed more difficult cells to receive more weight. The validity gate passed with 19/20 eligible trajectories.

The bounded support subset is defined before outcome analysis as $S = \{U_i : \max |U_i^{\text{normalized}}| \leq 3\}$. It removes proposals at the extreme tails of the normalized action distribution and re-runs the same selector comparison on the remaining candidates. Exact duplicates are retained in the inventory but deduplicated for strict winner selection; ambiguous P/O minima or negligible physical outcome ranges trigger abstention.

The primary recovery gate requires a positive O -versus- P effect under the preregistered trajectory-clustered test. A separate usefulness gate compares P against the exact expected outcome of uniformly sampling the same menus. Standard menu-level ranking correlations, pairwise accuracies, normalized regret, score decomposition, and model-specific summaries are descriptive: they explain the primary result but do not create additional confirmatory hypotheses.

C. Post-hoc selective diagnostic

After completing the preregistered analysis, we asked whether a planner could identify decisions that deserve additional verification using predictions alone. After collapsing physically duplicate actions, let $P_{(1)}$ and $P_{(2)}$ be the two lowest predicted costs in a menu. We define

$$m = \frac{P_{(2)} - P_{(1)}}{\max(\text{IQR}(P), 10^{-12})}, \quad (5)$$

$$U = -\log_{10}(\max(m, 10^{-12})),$$

so larger U means a closer predicted race. This score uses no realized latent or physical outcome. We evaluate it out of sample with leave-one-trajectory-out (LOTO) logistic prediction of the event $p \neq o$, including model identity as a baseline covariate. Each fold also sets a model-specific top-quartile threshold using only its 19 training trajectories. Uncertainty intervals resample whole trajectories 10,000 times. This analysis is exploratory and was specified in a time-stamped post-hoc addendum before its definitive run; it does not alter any preregistered test. The intervals condition on the chosen score and fixed out-of-fold predictions; they neither account for the earlier post-hoc choice of U nor refit the full cross-fitting pipeline inside each bootstrap resample.

a) Execution accounting.: Of 600 issued model rows, 577 produced merge-valid artifacts and 23 remained terminal-invalid; the stricter model-matched mask contained 562 rows (281 cells). Before outcome inspection, an outcome-blind infrastructure-recovery amendment was applied without changing candidate menus or the frozen analysis. Because this departed from the locked no-retry rule, we treat

the evidence as recovery-augmented and report an original-completions-only sensitivity below; complete attempt-level provenance is preserved in a sealed audit package.

V. RESULTS

Headline separation. Confirmatory effects use the 562 model-matched rows, whereas switch-only diagnostics use all 577 merge-valid artifacts. Predicted scoring beat uniform selection, but realized-endpoint scoring did not reliably beat predicted scoring. Thus, the score contained useful decision signal even though removing terminal prediction error did not by itself recover better choices.

a) Latent scoring contains action-selection signal.: P exceeded the exact expected physical outcome of uniformly sampling the same candidate menus by 48.70 in G (95% CI [40.47, 56.43], adjusted $p = 7.63 \times 10^{-6}$) and 0.761 in J ([0.420, 1.117], $p = 4.02 \times 10^{-4}$). Both outcomes passed the preregistered usefulness gate.

b) Realized endpoints do not reliably improve selection.: The raw $P \rightarrow O$ physical-outcome effect was -5.30 in G ($[-9.64, -0.71]$, adjusted $p = .0755$) and $+0.042$ in J ($[-0.176, 0.240]$, $p = .701$). Bounded- S estimates had the same qualitative pattern: -5.80 in G ($[-11.00, -0.21]$, $p = .111$) and $+0.115$ in J ($[-0.093, 0.314]$, $p = .296$). Although both marginal bootstrap intervals for G excluded zero, neither Holm-adjusted exact test passed $\alpha = .05$. Under the preregistered rule, the evidence does not support physical recovery. The interval and adjusted p -value are not contradictory: the percentile interval summarizes one marginal effect, whereas the Holm-adjusted exact test is the preregistered multiple-testing decision rule; the two need not cross their thresholds together. The G point estimate is 10.9% of the 48.70 predicted-versus-uniform advantage in magnitude, in the same simulator state-distance units. Descriptively, JEPA-WM accounts for most of the aggregate G difference (-9.33 , versus -1.27 for DINO-WM); no model-interaction test was preregistered, so this split is hypothesis-generating rather than confirmatory.

Most G cluster effects in Figure 2 are negative, with two large positive exceptions; J improvements and degradations are nearly balanced. This heterogeneity supports trajectory-level inference and argues against treating 562 model rows as independent samples.

c) Original-completions-only sensitivity.: As an exploratory check, we discarded every recovery-sourced artifact and reapplied the same fixed 15-cell matched aggregation. The remaining 408 matched rows yielded $\Delta^G = -5.39$ ($[-9.32, -1.40]$) and $\Delta^J = -0.019$ ($[-0.212, 0.160]$); bounded- S estimates were -6.30 and 0.052 . Only 14 trajectories met the original eligibility threshold, so this subset fails the locked validity gate and is not confirmatory. Its similar scale and the same substantive non-recovery conclusion suggest that the headline is not created by including the recovered rows.

d) Post-hoc latent interpolation.: To test whether full endpoint substitution created an isolated selector artifact, we scored stored interpolants $\hat{z}_\lambda = (1 - \lambda)\hat{z} + \lambda z$ for

TABLE I

TRAJECTORY-CLUSTERED PHYSICAL EFFECTS. POSITIVE FAVORS THE FIRST NAMED SELECTOR. PREDICTOR-SPECIFIC ROWS ARE DESCRIPTIVE BECAUSE NO MODEL-INTERACTION TEST WAS PREREGISTERED. p_H DENOTES THE HOLM-ADJUSTED EXACT SIGN-FLIP VALUE.

Contrast	Δ^G [95% CI]; p_H	Δ^J [95% CI]; p_H
O vs. P (raw)	-5.30 [-9.64, -0.71]; .0755	0.042 [-0.176, 0.240]; .701
O vs. P (bounded)	-5.80 [-11.00, -0.21]; .111	0.115 [-0.093, 0.314]; .296
P vs. uniform	48.70 [40.47, 56.43]; 7.63×10^{-6}	0.761 [0.420, 1.117]; 4.02×10^{-4}
DINO-WM: O vs. P (raw)	-1.27 [-7.84, 6.34]; -	0.003 [-0.243, 0.240]; -
JEPA-WM: O vs. P (raw)	-9.33 [-14.61, -4.67]; -	0.081 [-0.135, 0.279]; -

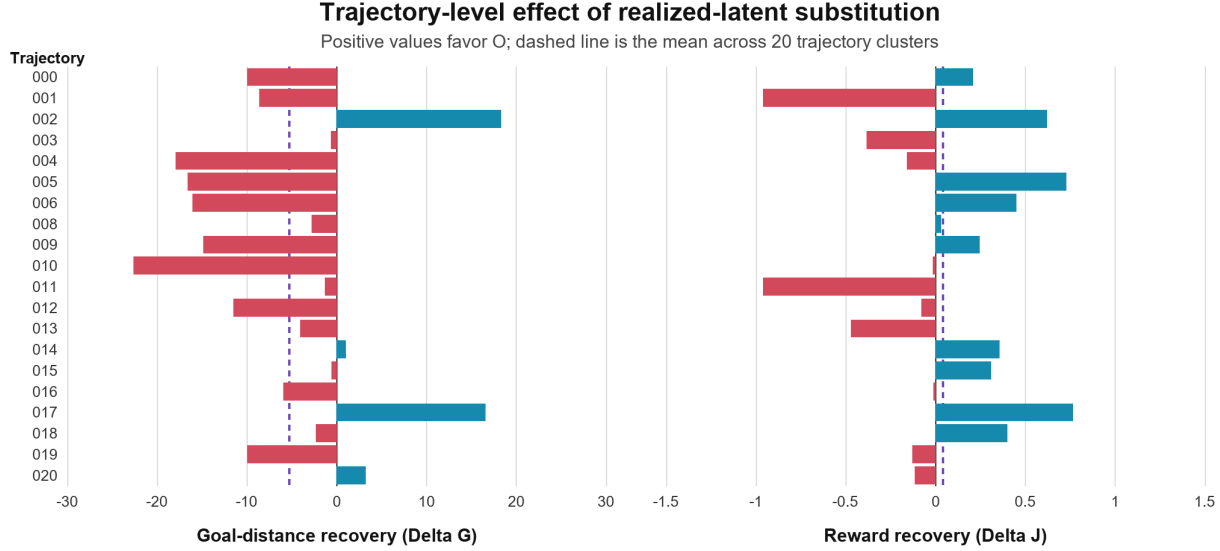


Fig. 2. The primary effect resolved by inferential unit. Realized-latent substitution improves goal distance on four of 20 trajectory clusters and reward on ten, but the magnitudes are heterogeneous. The dashed line is the unweighted cluster mean used in the raw analysis; this display is descriptive and does not replace the preregistered exact tests. All 20 scheduled clusters remain in the fixed-denominator estimate; eligibility is a validity threshold, not an exclusion rule.

$\lambda \in \{0, .1, \dots, 1\}$ without new simulator rollouts. At $\lambda = .5$, 38.4% of selectors had changed and $\Delta^G = -2.24$; at $\lambda = 1$, these values were 58.9% and -5.30 . All ten nonzero- λ Δ^G estimates were negative, with a lambda-effect correlation of $-.946$; the endpoint was also negative for each model and in the original-completions-only sensitivity. Reward showed no graded trend ($r = .245$). This aggregate pattern argues against the full- O endpoint alone creating the result, but interpolated latents need not correspond to realizable states and the analysis is explicitly post-hoc.

e) Correction frequently reranks actions, but seldom recovers the oracle-best pair.: Across 577 merge-valid diagnostic artifacts, P and O selected different candidates in 337 rows (58.4%). Among those disagreements, 48 (14.2%), 38 (11.3%), and 11 (3.3%) produced recovery-aligned reversals of the strict P -versus- G -best, J -best, and both pairs, respectively. These pair counts deliberately cover only switches involving the strict oracle-best candidate; they are a diagnostic complement to the aggregate recovery test.

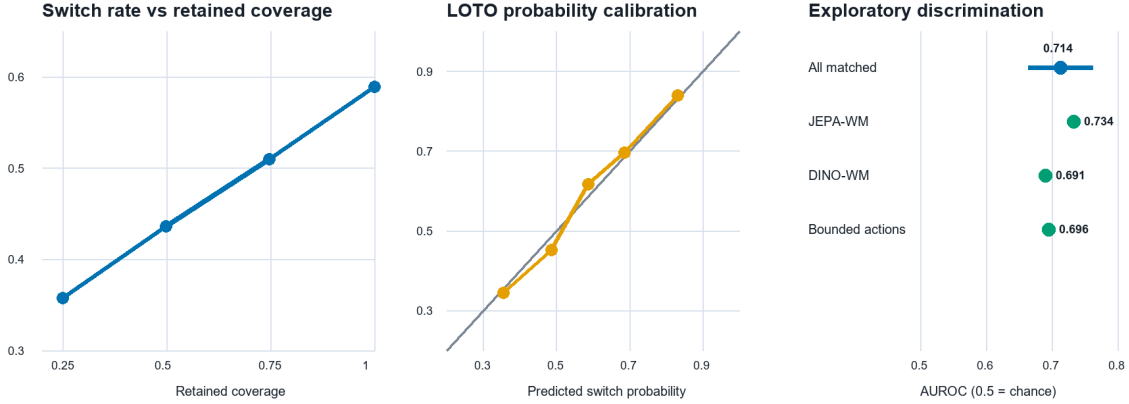
f) Prediction-only near ties flag selector instability.: On the 562 model-matched rows, U had exploratory AUROC .714 for P/O selector changes (trajectory-bootstrap 95% CI

[.665, .762]). The fitted LOTO probability model had AUROC .706. The training-fold-defined ambiguity quarter covered 25.3% of held-out rows and had 82.4% switch risk, compared with 51.0% elsewhere (risk ratio 1.62, [1.44, 1.82]). Point estimates remained above chance for DINO-WM (AUROC .691), JEPA-WM (.734), the bounded action set (.696), original completions only (.709; 408 rows across 15 observed trajectories), and all merge-valid artifacts (.714). The LOTO probability model improved Brier score from .243 using model identity alone to .214 with U (bootstrap difference $[-.043, -.016]$); raw and IQR-normalized margins were not reliably different.

Crucially, this is not a planner-regret result. The flagged-minus-unflagged normalized regret was $-.007$ for G (95% CI $[-.039, .026]$) and $-.056$ for J ($[-.143, .026]$). Near ties reveal that the selected index is sensitive to terminal prediction correction, but do not show that the deployed P action is physically worse. The score is therefore only a candidate signal for testing a future verification policy; no control, safety, or cost-benefit improvement was evaluated here.

Prediction-only near ties expose latent selector instability

562 model-matched decisions · trajectory-clustered inference · LOTO probabilities and thresholds



Top held-out ambiguity quarter: 82.4% switch risk vs 51.0% otherwise (risk ratio 1.62, 95% CI [1.44, 1.82]).

Interpretation: useful for flagging selector instability; it did not predict higher planner regret and is not epistemic/aleatoric uncertainty.

Fig. 3. Exploratory held-out selective diagnostic. Retaining increasingly ambiguous decisions raises the P/O switch rate (left); descriptive LOTO calibration is shown in the middle; discrimination point estimates remain above chance across both predictors and bounded actions (right). The near-tie signal predicts selector instability, not physical decision quality.

TABLE II

DESCRIPTIVE MENU-LEVEL RANKING DIAGNOSTICS ON THE PREREGISTERED BOUNDED SUPPORT S , AVERAGED OVER VALID ROWS. HIGHER IS BETTER FOR SPEARMAN AND PAIRWISE ACCURACY; LOWER IS BETTER FOR NORMALIZED REGRET@1.

Physical outcome	Diagnostic	Predicted P	Realized O
Goal distance G	Spearman rank correlation	.367	.345
	Pairwise ranking accuracy	.599	.591
	Normalized regret@1	.250	.274
Reward J	Spearman rank correlation	.121	.124
	Pairwise ranking accuracy	.469	.471
	Normalized regret@1	.372	.359

g) Prediction correction does not repair global ranking fidelity. The aggregate result is also visible beyond the selected top action. Substituting O for P slightly reduces both goal-distance ranking measures and increases normalized G regret@1. Reward ranking is essentially unchanged, with small improvements in pairwise accuracy, correlation, and regret@1. Therefore the absent physical recovery is not merely an argmin anomaly: under this fixed latent objective, more faithful terminal inputs do not consistently produce a more physically faithful ordering of the 300 candidates. Candidate support was not empty: the descriptive *oracle-within-top-k* upper bound, which uses outcomes and is not an executable selector, reduced normalized regret from .250 to .101 for G and from .372 to .224 for J when $k = 5$; $k = 10$ reduced it further to .064 and .179. The predictor often placed useful actions near the top without reliably identifying the best one.

h) The exact decomposition audits score changes, not physical causes. Equation 3 reconstructed $P - O$ to numerical precision (maximum identity residual 6.66×10^{-16}). Directional terms dominated magnitude terms in 266 of the 337 selector switches (78.9%), but this label frequency was nearly unrelated to trajectory-level recovery. The decomposition can therefore identify which algebraic term flips a latent ranking; in this study it does not establish a stable physical failure mechanism.

VI. DISCUSSION

a) Central finding. Taken together, the results reveal a gap between useful latent scores and physically aligned decisions. Predicted scoring was substantially better than uniform selection, so the planning signal was not arbitrary. Yet scoring the realized endpoints changed more than half of the selected actions without producing reliable aggregate physical improvement. Under these frozen menus and the unchanged scoring interface, we therefore find no evidence that terminal prediction correction is sufficient to recover decision quality. Possible remaining sources include the representation, latent goal objective, terminal-only scoring, and candidate support. Related work studies these design choices directly [7], [8]; our contribution is a matched intervention that measures their combined consequence at the decision boundary.

b) Using the audit as a diagnostic tool. The audit is designed for a released planner rather than a particular training algorithm. It produces five linked outputs: (i) P/O selector agreement, (ii) native regret recovered or added

by a switch, (iii) whole-menu ranking fidelity, (iv) strict oracle-pair reversals, and (v) exact D/M attribution of the latent score change. Together these distinguish “the prediction changed” from “the ranking changed” and “the physical choice improved.” A practitioner can run the same trace before retraining: strong O -over- P recovery implicates terminal prediction error as a recoverable bottleneck, while persistent O regret directs attention to the objective, representation, temporal score, or proposal set. That routing is diagnostic, not causal identification; the latter requires a second intervention on the suspected component.

The post-hoc near-tie result adds an online-facing output to this offline audit. A planner can compute U from its existing predicted costs. Our result motivates testing whether using it to route ambiguous menus to a more expensive verifier, fallback, or human improves utility. Because U did not identify higher physical regret, this paper does not claim that such routing improves control, safety, or cost-benefit tradeoffs.

VII. LIMITATIONS AND REPRODUCIBILITY

The study is limited to one simulated task and 20 independent trajectory clusters. Its two predictors share DINOv2 visual features, so the descriptive model split cannot support a general JEPa-versus-DINO conclusion. The audit uses six model actions (five simulator steps each), terminal-only scoring, and iteration-0 300-action menus rather than the released planner’s complete 30-iteration CEM or closed-loop execution. A finite menu can also hide recovery when it contains no materially better candidate, although the top- k regret results show that better actions were commonly present near the predicted top.

O uses simulator-privileged endpoints and is an evaluation intervention, not an online policy. The fixed-denominator rule avoids differential model exposure but shrinks missing cells toward zero; because that shrinkage can favor a non-recovery conclusion, the original-completions-only sensitivity is reported separately. Failures correlated with decision difficulty could still affect the estimand. No independent replication was run. The selective analysis was designed after observing the main non-recovery result and must be treated as hypothesis-generating despite LOTO evaluation and trajectory-level intervals. Upon acceptance, we will release an anonymized audit package containing the frozen manifest, preregistration, recovery addendum, issuance and attempt ledgers, merge index, sealed result, analysis code, and deterministic figure inputs. The 241-GB per-row arrays are not duplicated in that package; their paths and cryptographic bindings remain in the manifest and merge index for full reconstruction from the retained evidence.

VIII. CONCLUSION

On Push-T, the audited latent scores selected better actions than a uniform baseline, but replacing predicted endpoints with realized endpoints did not yield statistically reliable physical improvement. The same correction changed 58.4%

of selections while leaving whole-menu physical ranking fidelity nearly unchanged. These results illustrate why evaluating a world model for planning requires more than measuring prediction error: the relevant question is whether predicted futures preserve the action rankings that matter physically. The $P/O/G/J$ audit turns that question into a matched, reproducible test without retraining the model. A prediction-only near-tie score can additionally triage which choices are likely to change under endpoint correction, but a separate physical-risk signal is needed before such triage can promise better control.

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